Lab 1: Requirements Engineering & UML Use-Case Modelling

Name: Pranav Prabhakar
SRN: PES1UG24AM463
Class: 5-H

Requirements Specification

Problem Statement #61: Household Carbon Footprint & Target Tracker

A green living analytics tool where households log monthly utility consumption (electricity in kWh, cooking gas, commute km) to compute CO_2 equivalents and track reduction milestones.

Actors: Resident (primary), Sustainability Coach (primary), Emission Factor Service (supporting system actor)

---

1. Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest a household's monthly electricity (kWh), cooking gas (m^3 or kg LPG) and commute distance (km, by transport mode) logs, and shall compute the total monthly carbon equivalent (CO_2e) itemised by emission source.	High	**Pass:** Computed CO_2e for a reference dataset matches the EPA conversion factor formula within ±0.5%, and each of the three sources appears as a separate line item. **Fail:** Output is zero, negative, or omits a logged source.	The footprint value is the main output of the system. Every milestone, chart and recommendation depends on it being correct.
FR-002	Functional	The system shall validate every consumption entry before saving it. It shall reject values that are not numeric, are negative, or fall outside a plausible range. It shall block a duplicate entry for a household, source and month that is already logged, and shall require a billing month and year.	High	**Pass:** Entries of `-50` kWh, `"abc"`, and a second January 2026 electricity record are each rejected with a message on the field, and nothing is written to the database. **Fail:** Any invalid or duplicate record is saved, or quietly overwrites an existing one.	Invalid readings spread into footprint totals and trend charts. They are much harder to find and correct there than at the point of entry.
FR-003	Functional	The system shall allow a Resident to define a reduction target, either as a percentage or as an absolute reduction in kg CO_2e, against a chosen baseline period and target date. The system shall derive the monthly milestones from that target.	Medium	**Pass:** A 20% target over 12 months from a 500 kg CO_2e baseline generates 12 milestones whose allowances fall steadily and add up to the stated target. A target with a date in the past is refused. **Fail:** Milestones are not generated, or the arithmetic does not match the baseline.	Emission totals on their own do not change behaviour. Dated milestones are what make the tool a tracker rather than a calculator.
FR-004	Functional	The system shall compare each month's computed CO_2e against the active milestone, classify the household as On Track, At Risk or Achieved, and notify the Resident within 24 hours of computation. It shall also notify the linked Coach where consent exists.	High	**Pass:** A milestone that is met gives the status Achieved, plus a notification that is sent and logged. A milestone that is exceeded shows At Risk on the dashboard. **Fail:** The status is wrong, or no notification record is created within 24 hours.	This closes the loop between action and outcome. Feedback that arrives on time is what keeps a resident engaged from one month to the next.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall allow a Sustainability Coach to view historical footprint breakdowns and to attach dated written recommendations that the Resident can see. This shall apply only to households that have granted consent.	Medium	**Pass:** A coach with consent can read the breakdown and save a recommendation that the Resident can then see. A coach without consent receives HTTP 403 and the attempt is written to the audit log. **Fail:** Any household data is readable by a coach without a recorded grant of consent.	This gives the Sustainability Coach something useful to work with, while the household stays in control of who sees its consumption data.

2. Nonfunctional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional (Performance)	The annual carbon reduction milestone chart shall render its interactive historical comparison graph, covering up to 36 months of data, in under 200 ms at the 95th percentile, with at least 500 users active at the same time.	High	**Pass:** A k6 or JMeter benchmark at 500 concurrent sessions reports a 95th percentile render time of 200 ms or less, with an error rate below 1%. **Fail:** The 95th percentile exceeds 200 ms, or the error rate exceeds 1%.	The dashboard is the screen a resident comes back to each month. Visible lag on the main view is the quickest way to lose them, so the limit is set now rather than tuned later.
NFR-002	Nonfunctional (Security and Privacy)	All household consumption and identity data shall be encrypted in transit using TLS 1.3 and at rest using AES 256. Access shall be granted by role. Every coach read and every data export shall be written to an audit log that detects tampering and is kept for 12 months.	High	**Pass:** A security scan confirms that no channel carries plaintext and that the required ciphers are configured. Access tests show that reads across households are denied. Audit entries exist for 100% of sampled coach reads and exports. **Fail:** Any plaintext channel, any successful read across households, or any missing audit entry.	Monthly utility and commute data reveals when a home is occupied and what the daily routine looks like. That makes it sensitive personal data, even though each reading on its own looks ordinary.

3. Traceability

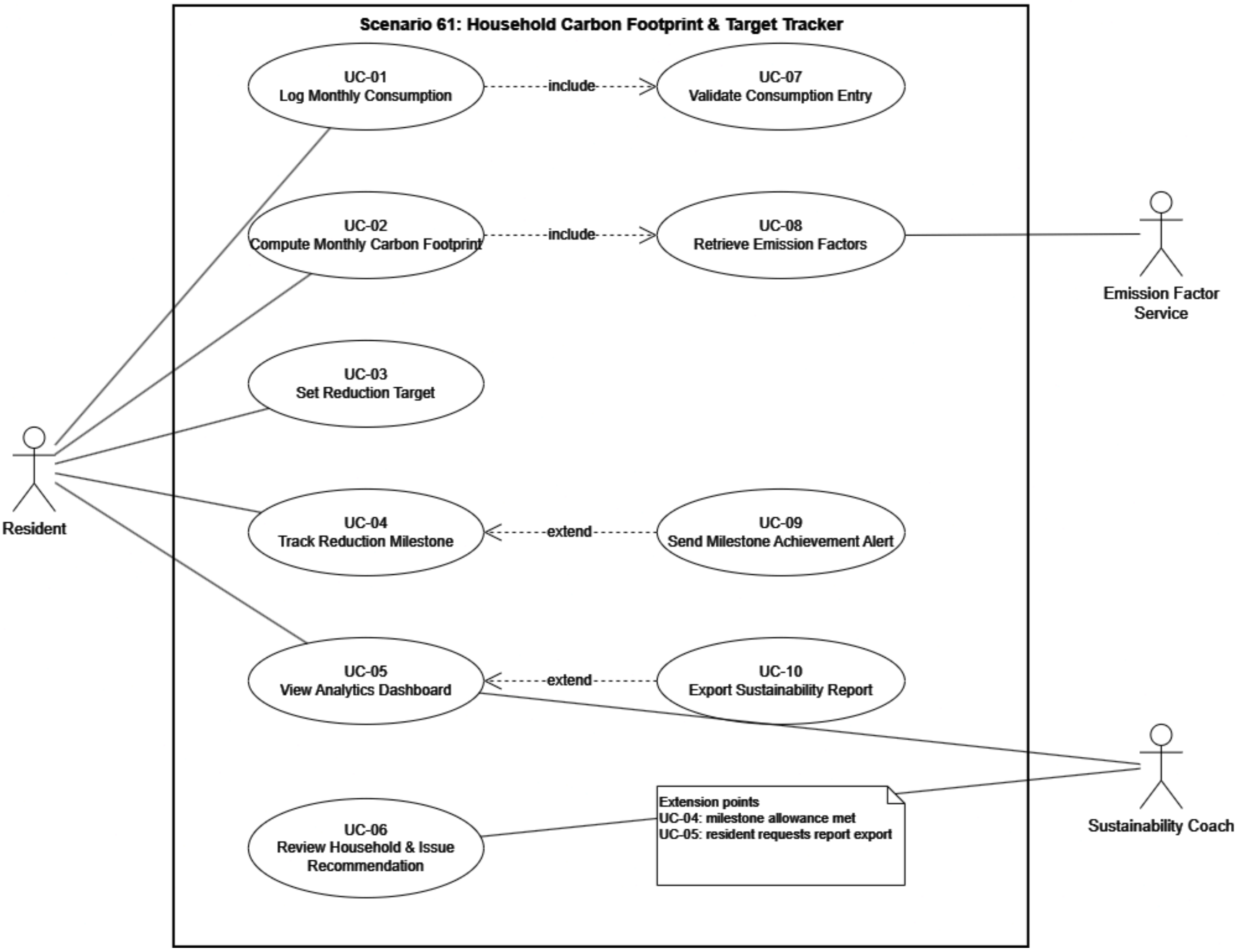
Requirement	Realised by
FR-001	UC-02 Compute Monthly Carbon Footprint, UC-08 Retrieve Emission Factors
FR-002	UC-01 Log Monthly Consumption, UC-07 Validate Consumption Entry
FR-003	UC-03 Set Reduction Target
FR-004	UC-04 Track Reduction Milestone, UC-09 Send Milestone Achievement Alert
FR-005	UC-06 Review Household and Issue Recommendation
NFR-001	UC-05 View Analytics Dashboard
NFR-002	UC-05, UC-06, UC-10 (applies across all three)

Lab 1: Requirements Engineering & UML Use-Case Modelling

Name: Pranav Prabhakar
SRN: PES1UG24AM463
Class: 5-H

UML Use Case Diagram

Problem Statement #61: Household Carbon Footprint & Target Tracker



Lab 1: Requirements Engineering & UML Use-Case Modelling

Name: Pranav Prabhakar
SRN: PES1UG24AM463
Class: 5-H

Use Case Flow Specification

UC-02 Compute Monthly Carbon Footprint

Field	Value
Use Case ID	UC-02
Name	Compute Monthly Carbon Footprint
Primary Actor	Resident
Supporting Actor	Emission Factor Service (external system)
Trigger	The Resident submits or edits the last consumption entry for a billing month, or the scheduled month end job runs.
Includes	UC-08 Retrieve Emission Factors
Related Requirements	FR-001 (realised here), FR-004 (through the handoff at step 8), FR-002 (met earlier by UC-01, assumed by precondition 3)

Preconditions

1. The Resident is authenticated and belongs to the household in question.
2. The household profile is complete, including the grid region used to pick the electricity emission factor.
3. At least one **validated** consumption entry for electricity, gas or transport exists for the target month.
4. An emission factor set is available, either live from the Emission Factor Service or from a cached set that is still inside its published validity window.

Postconditions

On success

1. A monthly footprint record is saved. It holds the total CO_2e, a breakdown by source, the version of the emission factor set used, and a computation timestamp.
2. The milestone status for the household is recalculated, which feeds UC-04.
3. The analytics dashboard shows the new month the next time it loads.

On failure

1. No partial footprint record is written. The month stays in the state `PENDING_COMPUTATION`.
2. The reason for the failure is logged and shown to the Resident.

Main Success Scenario

Step	Actor	Action
1	Resident	Selects the billing month and asks for the household footprint to be computed.
2	System	Loads all validated consumption entries for that household and month, grouped by source.
3	System	Checks whether the month was already computed with the current factor set version. If it was, the run is flagged as a recomputation.
4	System	**«include» UC-08:** asks the Emission Factor Service for the factor set covering the household's grid region and its fuel and transport modes.
5	Emission Factor Service	Returns the current factor set with its version identifier and validity period.
6	System	Converts each entry to CO_2e and sums the results. Electricity kWh is multiplied by the grid factor, gas quantity by the fuel factor, and commute km by the factor for that mode.
7	System	Saves the footprint record with the breakdown by source, the factor set version and a timestamp.
8	System	Recalculates the milestone status against the active reduction target and hands off to UC-04.
9	System	Displays the itemised monthly footprint, compared against the previous month and against the milestone allowance.
10	Resident	Reviews the breakdown and closes the view.

Alternate Flow

A1: Emission Factor Service unavailable, branching at step 5

Step	Actor	Action
5a.1	System	Detects that the Emission Factor Service did not respond within the 3 second timeout, after two retries.
5a.2	System	Fetches the most recent cached factor set and checks it against its published validity period.

Step	Actor	Action
5a.3	System	**If the cached set is still valid:** resumes the main scenario at step 6, marks the resulting record `PROVISIONAL`, and records the cached factor set version.
5a.4	System	Displays the footprint with a notice that it was computed from cached factors and may be revised.
5a.5	System	Queues a recomputation for when the service is reachable. If the new total differs by more than 1%, the record is updated and the Resident is notified.
5a.6	System	**If no valid cached set exists:** stops the computation, leaves the month in `PENDING_COMPUTATION`, and tells the Resident that the footprint will be calculated once factor data is restored.

Postcondition of A1: Either a `PROVISIONAL` footprint record exists with a recomputation queued, or no record exists and the month stays pending. In neither case is a figure stored without a version or only partly computed.